HOMEWORK 2 EE4520 2025-2026

Due date: Sunday, February 1, 2026 (electonic submission, see Brightspace)

Introduction:

In this project you are going to design a discrete-time fully differential folded-cascode amplifier to certain given specifications. The specifications will be different for each student, although some parts of the specifications are in common. The common part is to design an OpAmp in a feedback situation (as depicted below in fig. 1), with a **closed-loop gain of 8x (Acl = 8)**, in 0.18 μm CMOS, operating from a 1.8 V supply. In fig. 1 the resistors are there *only for biasing purposes* and can be chosen very large: R_{in} = 100 GΩ and R_{fb} = $A_{cl}R_{in}$ = 800 GΩ. The ratio of the feedback capacitors needs to be chosen such that the closed-loop gain equals A_{cl}, or in other words: $C_{fb} = C_{in} / A_{cl}$. The load capacitance needs to be chosen equal to the (input) feedback capacitance: $C_{load} = C_{in}$. The absolute values of the capacitors need to be chosen by you, such that you achieve your design goals.

OpAmp schematic:

The OpAmp circuit diagram is shown in fig. 2. It is a folded-cascode OpAmp with ideal gain-boosting and ideal common-mode feedback. The implementation of the ideal gain-boosting blocks (nmos and pmos) is shown in figs. 3 and 4, whereas the ideal common-mode feedback is depicted in fig. 5. **It is mandatory to use these exact implementations.** For biasing you can use the biasing block you designed in the homework. Make sure to have the correct sizing of the biasing block checked by one of the TAs (you don't want an error in the biasing be the cause of problems in the design of the OpAmp). Use the same transistor-sizing as in your homework and also use the same unit W and L in this project. Use the m-factor to change transistor-sizing. **Please connect the back-gates for all nmos transistors to V_{ss} and for all pmos transistors to V_{dd} in all circuits.**

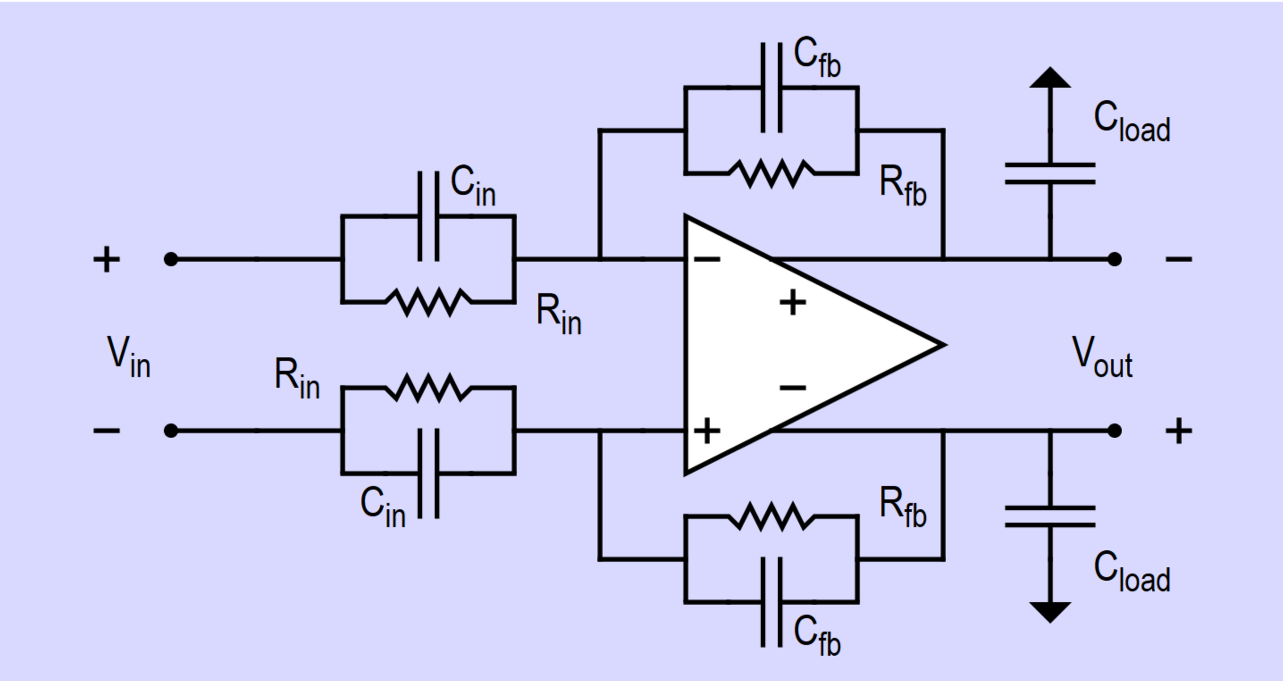

Fig. 1. OpAmp in feedback situation

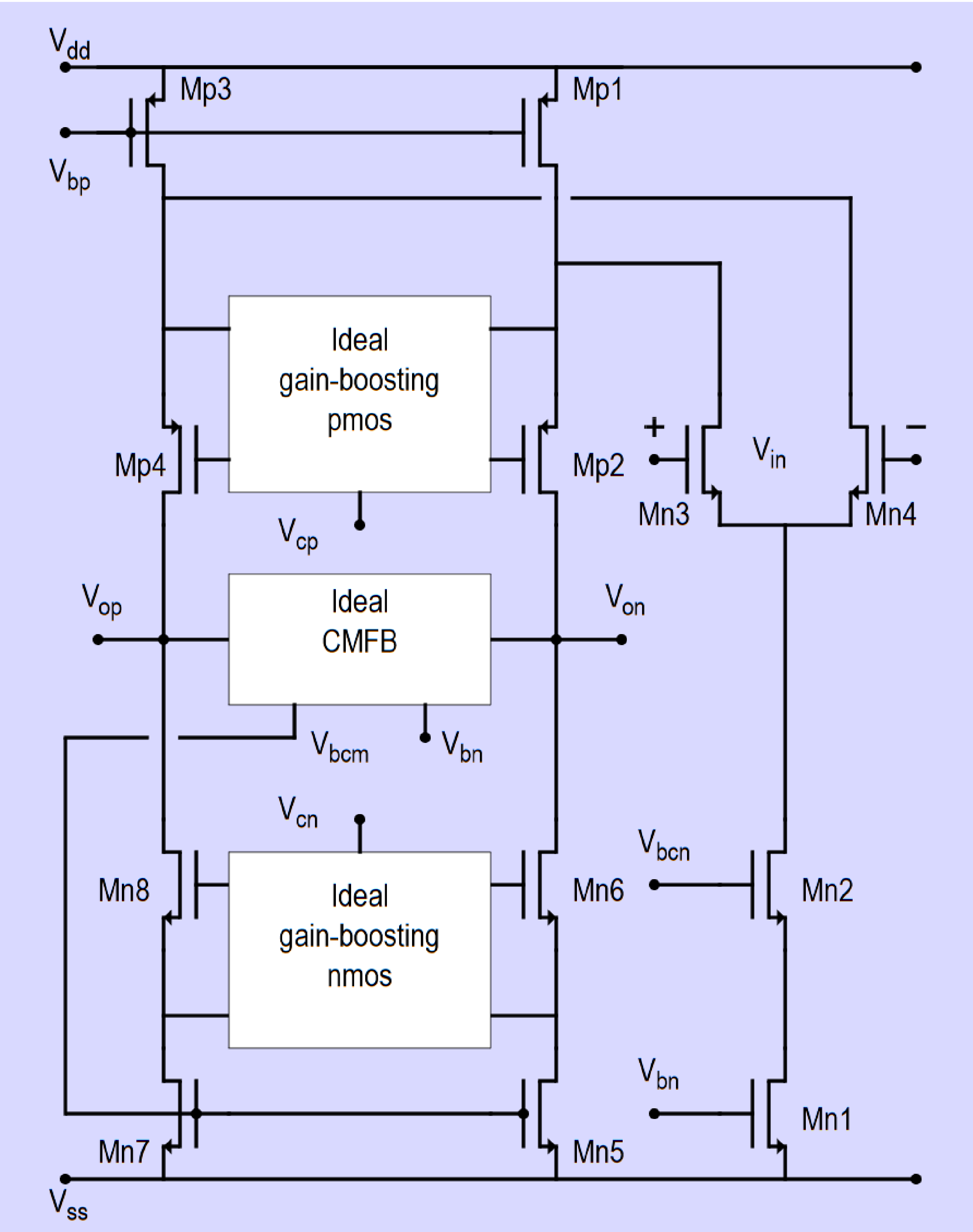

Fig. 2. OpAmp with ideal Gain-Boosting

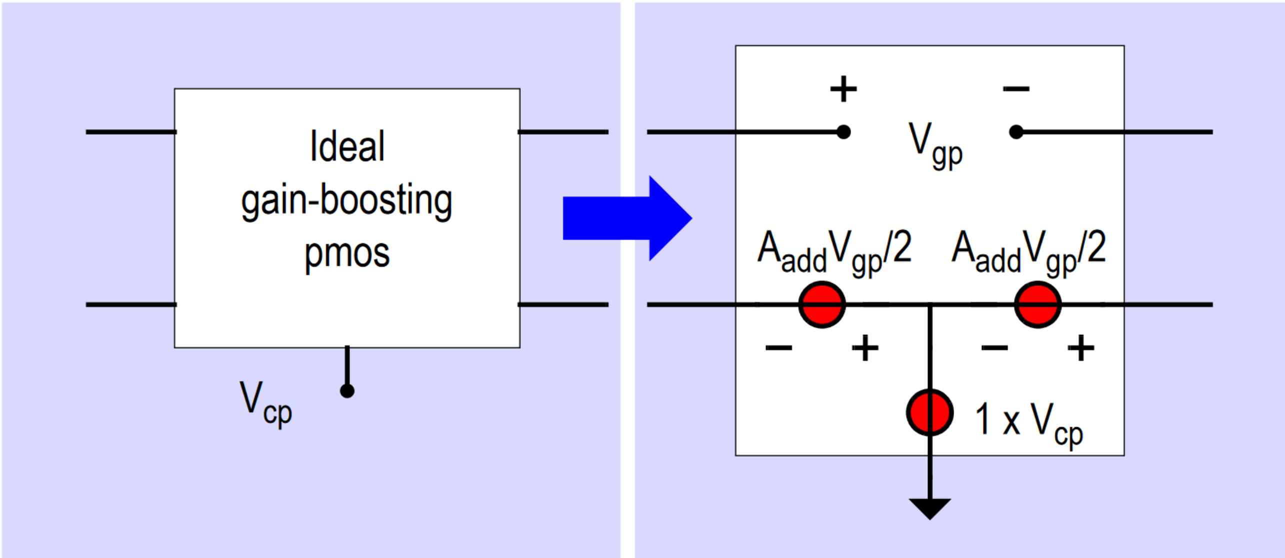

Fig. 3. Implementation of ideal Gain-Boosting pmos

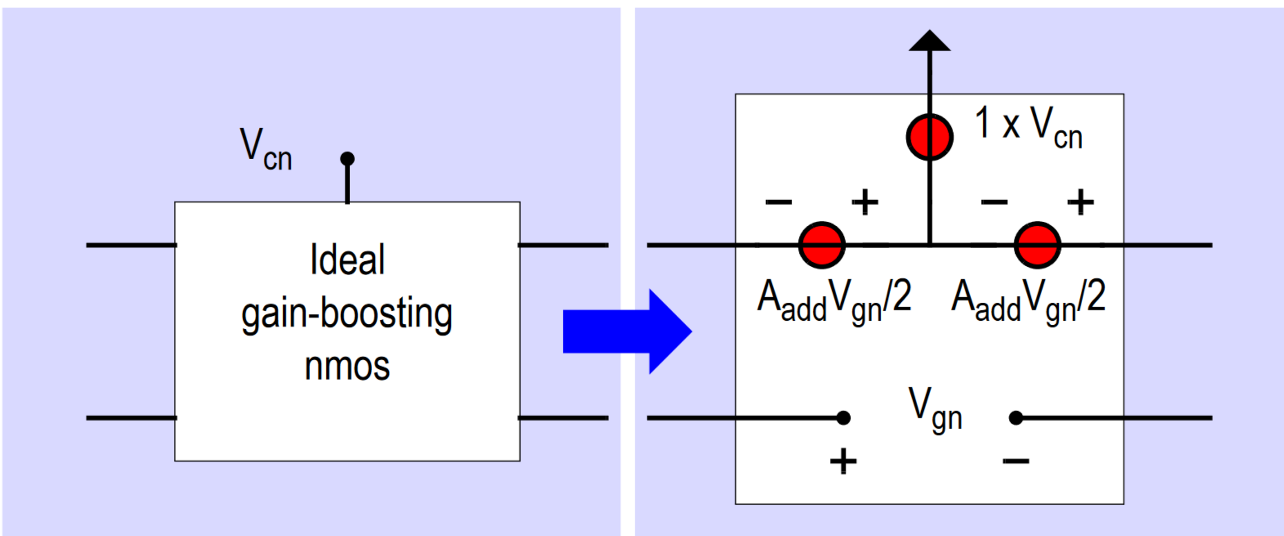

Fig. 4. Implementation of ideal Gain-Boosting nmos

Ideal Components:

The sources in red in figs. 3, 4 and 5 are voltage controlled voltage sources. They copy another voltage and multiply this with a gain (for instance $A_{add}/2$ or 1) to mimic the behavior of an amplifier or buffer (gain = 1). This allows for perfect control but is of course also a simplification of reality. This implementation of the amplifier does not show any frequency dependent behavior and does not require any design. The same is true for the ideal CMFB-circuit depicted in fig. 5. The source in red is also a voltage controlled amplifier. It is inserted to ensure that the CM-level of the output voltages V_{op} and V_{on} is at 0.9 V. The 2 resistors (R_{big}) are to sense the DC CM-output voltage and are to be chosen large enough such that their value does not reduce the circuit's output impedance. The capacitors C_{CM} are taking care of the CM-feedback for higher frequencies and have to be chosen equal to the feedback capacitance: **$C_{CM} = C_{fb} = C_{in}/8$** and **$C_{load} = C_{in}$**. Please use a value of **$A_{add} = 100x$** in this homework.

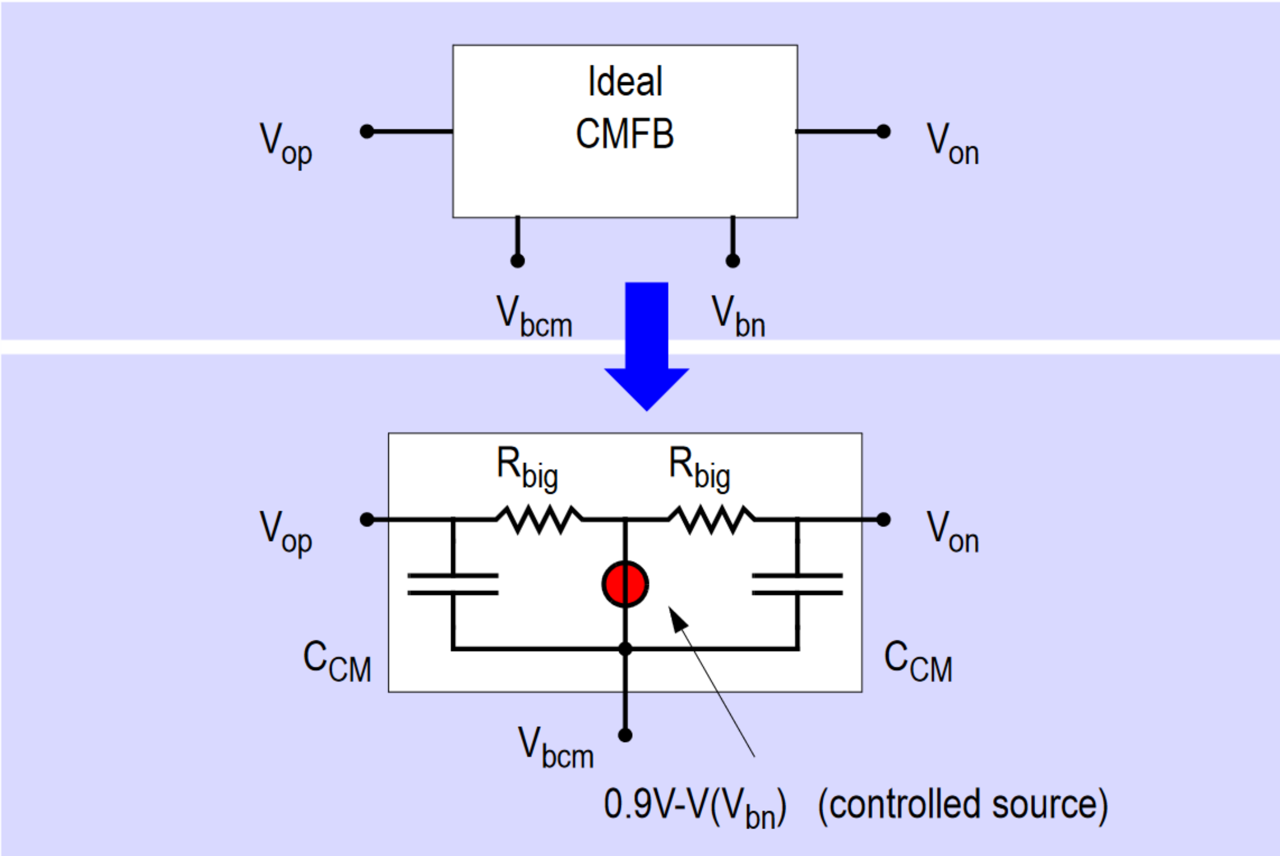

Fig. 5. Implementation of ideal CMFB

Types of simulations:

You are asked to run 3 types of simulations: ***transient***, ***AC*** and ***noise simulations***.

Transient simulations are required to show a step-response. As your OpAmp is intended for discrete-time applications, a step-response is sufficient to show its behavior with respect to speed and accuracy. The output-step should be **1.2 [V_{pd}]**.You will need to apply a step at the input of **(1.2/A_{cl}) [V_{pd}]** such that the output shows the required step. The required output step equates to a maximum signal of **0.8485 [V_{rms}]**. You will be given specifications for how many dBs you have to settle within what amount of time (this is different for every student).

The **AC simulations** should be run from **1 Hz to 100 GHz** and show a plots of the open-loop gains A and Aß versus frequency, with gain and phase margin indicated, as well as the unity-gain frequency. A second plot should show the closed-loop gain versus frequency, showing the -3 dB bandwidth.

The **noise simulation** needs to show the output noise (Onoise) and the integration of that between **10 kHz and 100 GHz (note that this is not the same frequency range as for the AC simulations - do not integrate from a lower frequency than 10 kHz as the model still has some artifacts of 1/f-noise at very low frequencies)**. It is important to use the correct start and stop-frequency for the noise integration, as it will influence the outcome of the total integrated noise. For ease of design and for easy comparison, we will not include 1/f-noise in this homework. A special parameter-set for the 0.18 µm process (***'logic018 no1of.l'***) will be supplied that does not include 1/f-noise (this file uses the file: ***'log018 Delft no1of.txt'***, please make sure you have both files in your working directory).

Please make sure you use the correct parameter set for this homework. The integrated value of the Onoise is given in μV_{rms} (or mV_{rms}). Comparing that to the maximum signal of 0.8485 [V_{rms}] will give you the SNR (either on a linear scale or in dBs). Each of you will be given a different value of SNR in dBs.

Goal:

The goal of this homework is to achieve all targets given to you with the lowest possible power dissipation. In your transient simulation you can plot the current drawn from V dd . That current (the **average**) multiplied times the value of **V_{dd} (1.8 [V])** yields the power dissipation of your design. Please use a **different supply for the biasing (for instance V_{ddb} = 1.8 [V])**, such that the current in V_{dd} ($I(V_{dd})$) only reflects the current drawn by your circuit and not from the biasing circuit. The goal is to achieve the specifications given to you (i.e. settling accuracy (A_{settle}), settling-time (T_{settle}) and SNR) using **the lowest possible power dissipation** (Power).

Your Personal Target Specifications:

Each student will get an unique set of specifications. To get your own set of specifications, follow the instructions posted on Brightspace together with this assignment. Your individual specifications have to be inserted in the ***Table of Results*** (Table 1 - see Deliverable 1) in rows 2, 3 and 4, in the column labelled 'Spec'. Please also insert the rest of your design results in the table, as shown in Table 1 (Deliverable 1). To accommodate for (minor) differences arising from the given specifications, bonus FoM points are given (row 22).

Note about the plots in this homework description:

With the exception of the circuit architectures, the plots shown in this homework description are all **examples only**. The numbers shown in the plots are **not** (necessarily) your target specifications. Even the frequency ranges, voltages, currents, settling accuracies, etc. serve as examples only and **are not your goal**. Your specifications are individually for you and the results that various students obtain will be different.

Minimum Target for Figure-of-Merit (FoM):

Most of the work in this homework will culminate in a single number: the Figure-of-Merit (FoM), as defined in the table (bottom rows). To give you a feeling for what number needs to be achieve, we give you a minimum target for the FoM. That minimum number is the same for all students and is: **FoM ≥ 174 [dB]**. The higher the number, the better the result, the higher your grade will be (assuming all tasks were done correctly). As an indication, the FoM of the best designs may exceed 176 dB.

Biasing:

It is very important to correctly bias your circuit. In homework 1 you have designed a biasing-block (fig. 6) which you can use perfectly to bias this amplifier. **You may have to adjust the voltages (currents) to get optimal results.** You can do that by controlling the bias currents $I_{b,main}$, $I_{b,pcas}$ and $I_{b,ncas}$. Please use the fixed relations :

- $I_{b,ncas}$ = $ratio_n$ $I_{b,main}$ and
- $I_{b,pcas}$ = $ratio_p$ $I_{b,main}$,

as determined in homework 1. If you have any doubts about your biasing circuit, please ask the TAs to check your circuit. You don't want your bias-block to be the source of trouble in this project. Do not forget to use a different supply (like V_{ddb}) for the baising circuit, so you can exclude the power dissipation in this circuit form the power dissipation in your design. Also, do not forget to use large de-coupling capacitors (~1μF) on all the bias lines. That way you can prevent that signals get on your bias-lines and affect other parts of the circuit. This should be a general habit for biasing blocks.

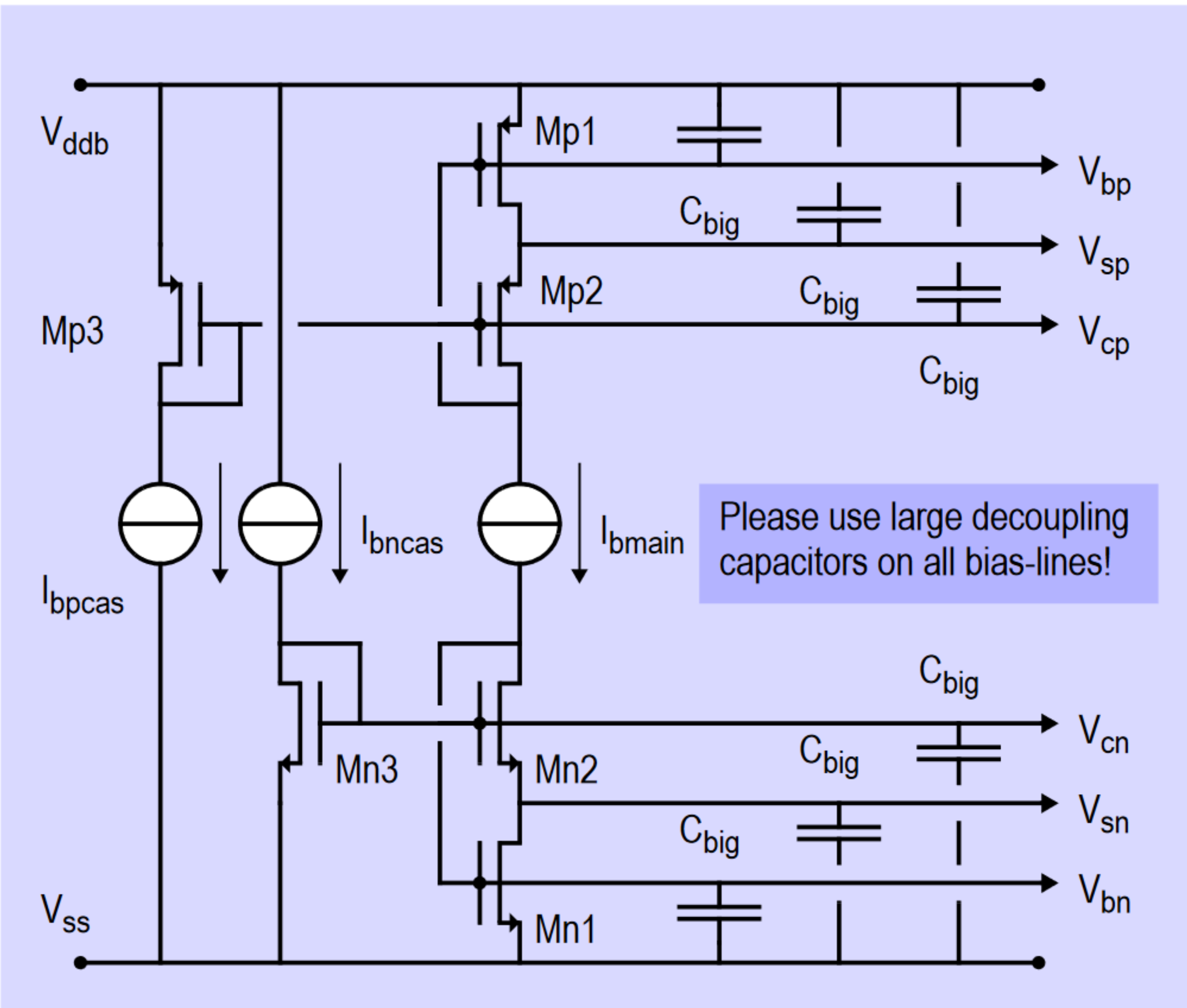

Fig. 6. Biasing circuit

Sizing of the Transistors:

Fig.2 shows the schematic of the OpAmp. In order to meet your design target, you need to properly size the transistors in this circuit. There is a lot of freedom in choosing sizes but there are a few limitations as well. Please make sure you stick to the following rules:

RULE #1:

First of all we'll ask you to use 'unit' transistor 'fingers' and do the final sizing using the 'm-parameter'. So we ask you to use the following:

W = 1.0µm (unit finger width)
L = 0.18µm (unit finger length)
m = free to choose

RULE #2:

To achieve a proper design, we also need to have a proper choice of the relative currents in the branches of the circuit of fig. 2. *In a proper design, the currents have to be chosen in such a way that if the differential pair (MN3,4)is completely tilted in one direction, the maximum current coming out of the differential pair does not exceed the available current coming from MP1,3*. If that would happen either MP2 or MP4 would turn off completely and we would lose the output branch. Therefore, the currents have to be chosen in the following ratios:

$$I_{d,MN1} = 2I_{d,MN5} = 2I_{d,MN7}$$

and

$$I_{d,MP1} = I_{d,MP3} = I_{d,MN1}$$

This forces the choice of m-parameters to be:

MN1,2 m = 4 * ScaleA
MN5,6 m = 2 * ScaleA
MN7,8 m = 2 * ScaleA

where ***ScaleA*** is a scaling parameter for the entire OpAmp and has to be chosen by you to meet all specifications. Similarly:

MP1,3 m = 4 * ScaleA * mp-ratio
MP2,4 m = 2 * ScaleA * mp-ratio

where the ***mp-ratio*** is the ratio between the N- and PMOS transistors to get the correct biasing determined in Homework 1. MN3,4 can be chosen freely, but we suggest to start with

MN3,4 m = 2 * ScaleA (suggestion)

IMPORTANT REMARK:

For this HW, you will be allowed to use non-integer (fractional) values for the number of fingers of a transistor, even though that this is not what would be used in practice of course.

Deliverables & Advice:

Deliverables:

You need to show the following information in your report:

1) Table of results, as shown in table 1.
2) Schematic with all node voltages, as shown in fig. 7 (not a simulator schematic).
3) Schematic with all branch currents, as shown in fig. 8 (not a simulator schematic).
4) Bode-diagram of A and Aß (including gain [dB] and phase), as shown in fig. 9. Please indicate BW_{cl} and τ_{cl}.
5) Bode-diagram of the closed-loop transfer-function, showing 8x gain (**18.06 dB**) (fig. 10). Please indicate the BW_{cl}.
6) Plot of settling accuracy versus time, showing the achieved accuracy at a given time (see fig. 11).
7) Plot of settling accuracy versus time, showing T_{settle} , T_{40dB} and $T_{48.69dB}$ (fig. 12).
8) Plot of output noise power density versus frequency (10 kHz – 100 GHz). Please make sure the integrated noise is shown (fig. 13).
9) Short description of your design process (max. 1 page, **20% of your <u>final</u> grade**). Explain how you arrived at the final biasing and sizing and the reasoning behind your design steps. Please also show how the results from the open loop gain A, loop gain Aß, closed-loop gain bandwidth and the settling-curve are linked.

Advice:

A) Please check with your **TAs for advice** regularly, it can save you a lot of time.
B) Before you start, have the **Biasing Circuit** (from the homework) checked. Be aware of the fact that, although you want to use the circuit architecture and sizing, you may want to use a different current density.
C) After that, have the simulation set-up for **Settling Simulations** and the simulation set-up for **Noise Simulations** checked as well.
D) You will be judged based upon how close you get to the results given to you. **Over-designing will not result in additional points**, but falling short of the specs will result in a penalty. Over-designing will cause a higher than necessary power dissipation and that will also show up in your grade. So, the goal is to get **as close as possible to the specifications given to you** while using as low as possible power dissipation.
E) Please note that for FoM_{dB} a compensation factor is given (row 22 in Table of Results - Deliverable 1). This is done to compensate for the (very) small differences in the resulting FoM based on the set of parameters you were given. This factor (which you add in dBs) will compensate for that. Note that the FoM_{dB} should be a positive number and the bonus should make the number higher (better).
F) Pay attention to the **compulsory order** in your report.

<u>DESCRIBE YOUR DESIGN PROCESS; IT COUNTS SIGNIFICANTLY TO YOUR FINAL GRADE!</u>

Example of deliverable 1:
Table of results

The shown numbers are *examples* only and not goal by itself!

Item	Design Item	Unit	Spec.	Achieved
1	Design Number (given to you)	Individual numb.		
2	SNR	[dB]	Individual spec.	71.62
3	A_{settle}	[dB]	Individual spec.	[illegible]
4	T_{settle}	[µs]	Individual spec.	1.30
5	BW_{cl} from open-loop AC simulations	[MHz]	---	0.597
6	BW_{cl} from open-loop AC simulations	[MHz]	---	[illegible]
7	Time-constant τ_{cl} from closed-loop AC simulations	[µs]	---	0.260
8	T_{40dB} (time to reach an accuracy of 40 dB)	[µs]	---	[illegible]
9	$T_{48.69dB}$ (time to reach an accuracy of 48.69 dB)	[µs]	---	[illegible]
10	Time-constant τ_{cl} from transient simulation	[µs]	---	0.266
11	Power dissipation	[µW]	Lowest possible	72.5
12	$I(V_{dd})$	[µW]	Lowest possible	[illegible]
13	Total integrated noise	[μV_{rms}]	---	224
14	Input step voltage	[mV]	150	150
15	Output step voltage	[V]	1.2	1.2
16	C_{in}	[pF]	---	4
17	C_{fb}	[pF]	---	[illegible]
18	C_{load}	[pF]	---	4
19	C_{CM}	[pF]	---	0.5
20	R_{big}	[GΩ]	Big	1
21	C_{big}	[nF]	Big	10
22	Bonus points on FoM_{dB}		Individual numb.	0.0
23	$FoM_{lin} = 2\pi * Power * T_{8.6dB} / SNR_{lin}^2$	[J]	≤ 3.98e-18	7.08e-18
24	$FoM_{dB} = -10 \log_{10}(FoM_{lin})$ (positive number)	[dB]	≥ 174.0	171.5

Example of deliverable 2:
Schematic with all node voltages

Make sure the schematic is well readable. A copy from the schematic entry usually is not well readable. Clearly indicate the voltages on each node. You don't have to include the gain-boosters or the ideal common-mode feedback.

Please look at the example below. **The shown numbers are *examples* only and not goal by itself!**

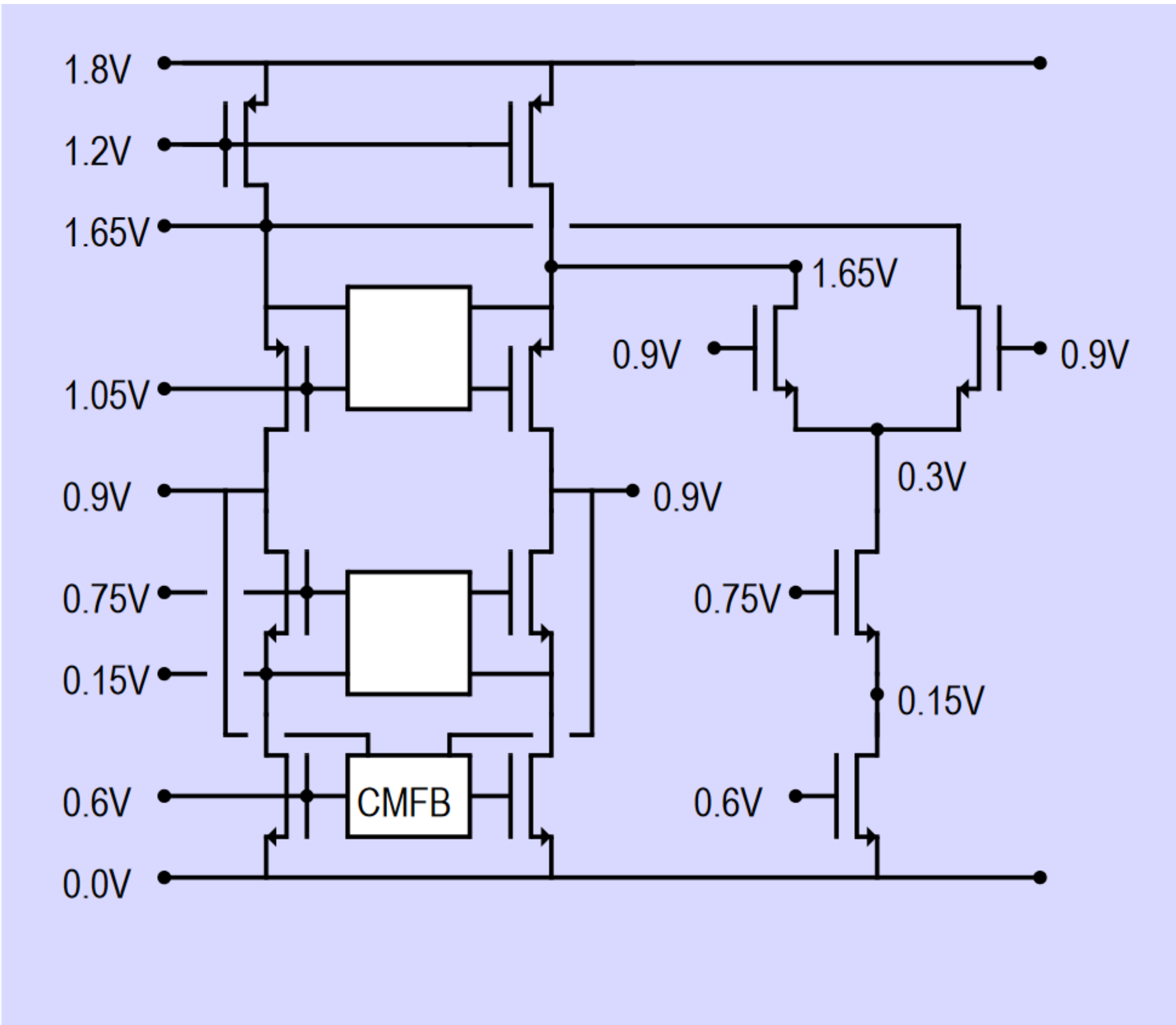

Fig.7. All node voltages

Example of deliverable 3:
Schematic with all branch currents

Make sure the schematic is well readable. A copy from the schematic entry usually is not well readable. Clearly indicate the voltages on each node. You don't have to include the gain-boosters or the ideal common-mode feedback.

Please look at the example below. **The shown numbers are *examples* only and not goal by itself!**

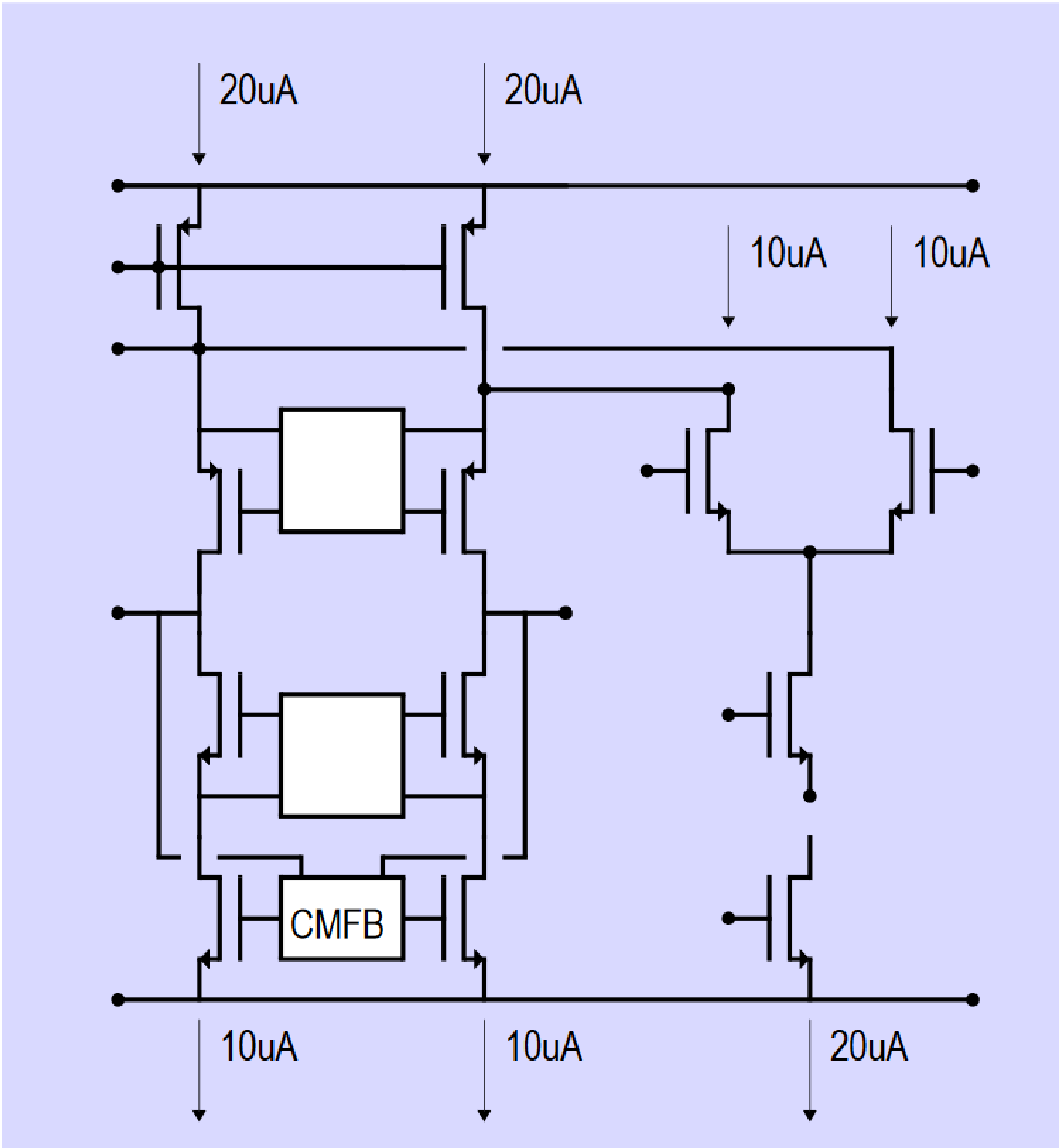

Fig.8. All branch currents

Example of deliverable 4:
Bode-diagram of A and Aß with Gain (dB) and Phase

Make a plot (Bode-diagram with gain and phase) of the amplifier gain A and the loop gain Aß versus frequency (**1 Hz – 100 GHz**), as shown below. Please indicate the gain- and phase-margin in the plot. Make sure you use the correct curve for that. Subsequently, use the unity-gain frequency (f_{unity}) and ß to determine the time-constant of the closed-loop system. Show the expected closed-loop bandwidth and time-constant in the graph. **The example below is just an example, the shown numbers are not a goal by itself (including the frequency range of the simulation)!**
Please use the targets given specifically to you.

When you open the loop for the Aß simulation, you can choose to have the ß-network either loaded by the OpAmp input or not. Please do both to compare the difference. **Explain what you see.** Please make sure that the OpAmp is in the same state is it normally is, i.e. the inputs are biased around the CM-level, the output is biased (or forced) around the CM-level and all the currents are identical to the normal situation (so the input parasitics are the same).

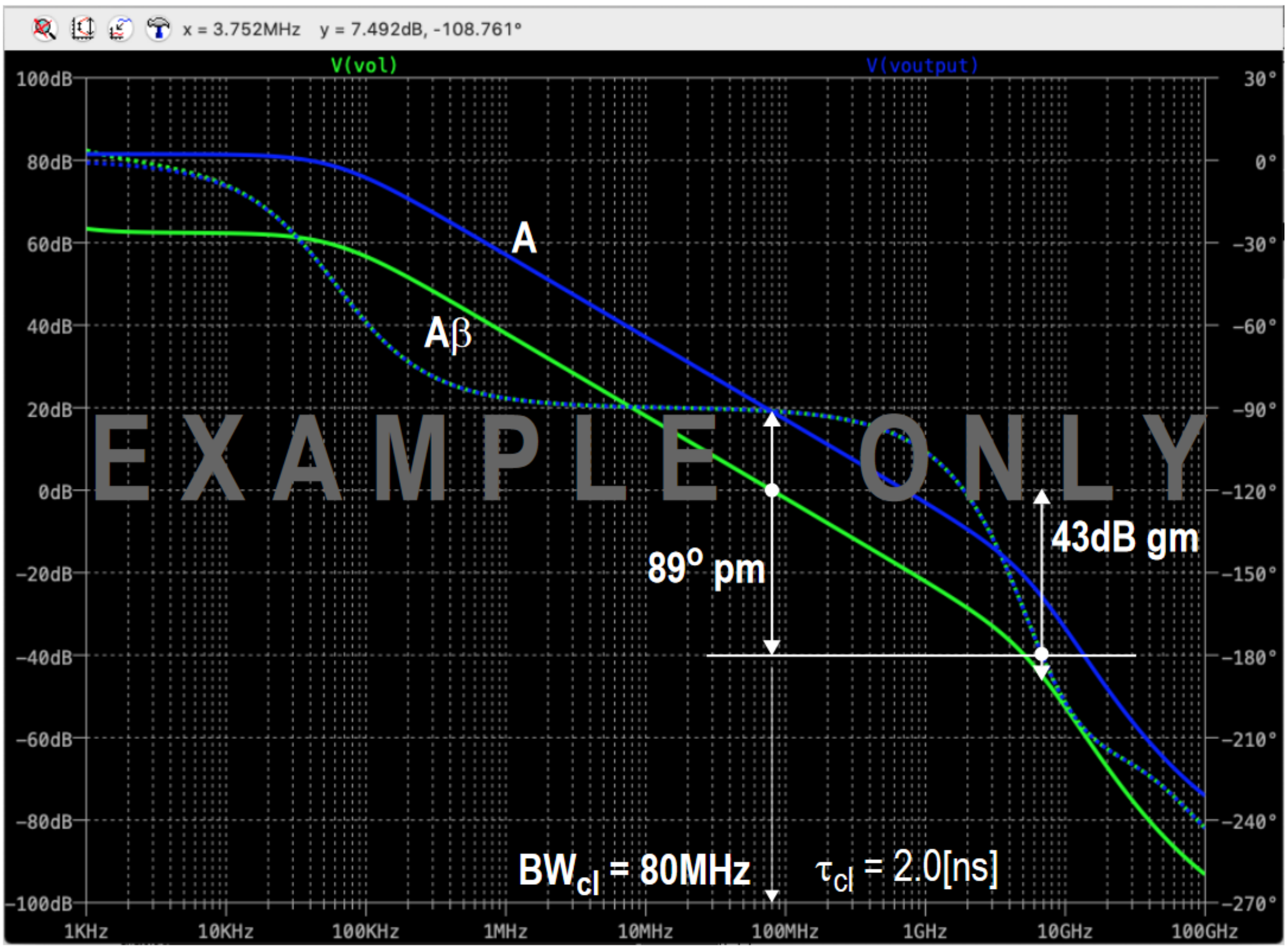

Fig.9. Bode-Diagram of A and Aβ.

Example of deliverable 5:
Bode-diagram of closed-loop Gain with Gain (dB) and Phase

Make a plot (Bode-diagram with gain and phase) of the closed-loop Gain of the system versus frequency (**1 Hz - 100 GHz**), as shown below. Please indicate the -3dB bandwidth and derive from that the closed-loop time-constant. **The example below is just an example, the shown numbers are not a goal by itself (including the frequency range of the simulation)!**

Make a comparison between the results you found in the open-loop simulation and the closed-loop simulation and explain any differences.

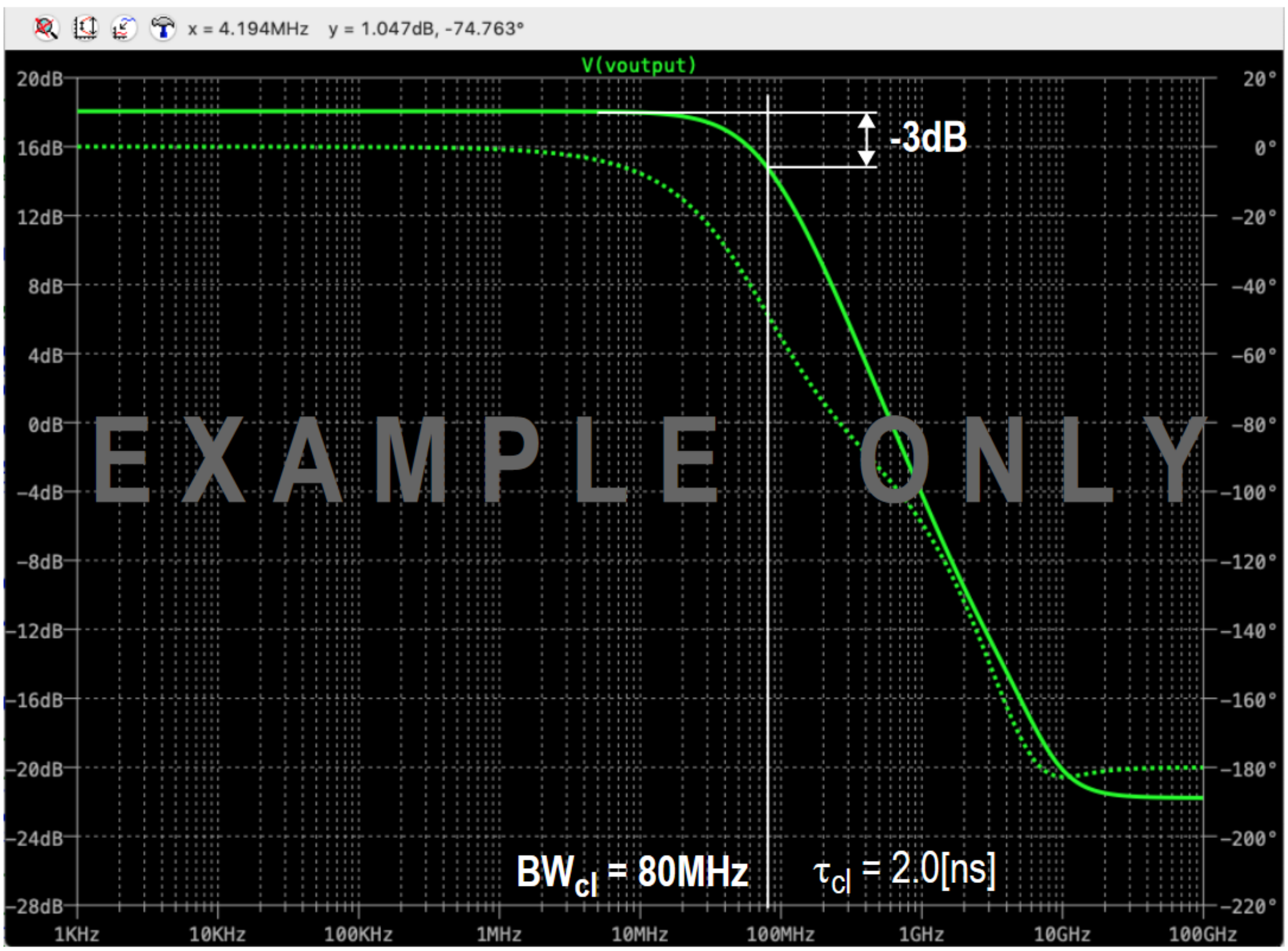

Fig.10. Bode-Diagram of closed-loop gain and phase.

Example of deliverable 6:
Settling accuracy versus time

In order to be able to judge the settling behavior, a plot of achieved settling accuracy versus time is needed, as shown below. The proper expression to be used for this plot is:

$$\text{Accuracy}_{dB} = 20 \cdot \log_{10}\left(\frac{1.2\ \text{V}}{\text{abs}(V(\text{virtualground})) + 1\text{n}}\right)$$

Please make sure you use the correct expression. Use a fixed 1.2 V as the upper range (not V(out) as a function of time). When in doubt, have it checked by a TA. The shape of the curve should be as depicted below. The value at the time given to you has to reach the accuracy given to you. Remember: over-designing costs power!

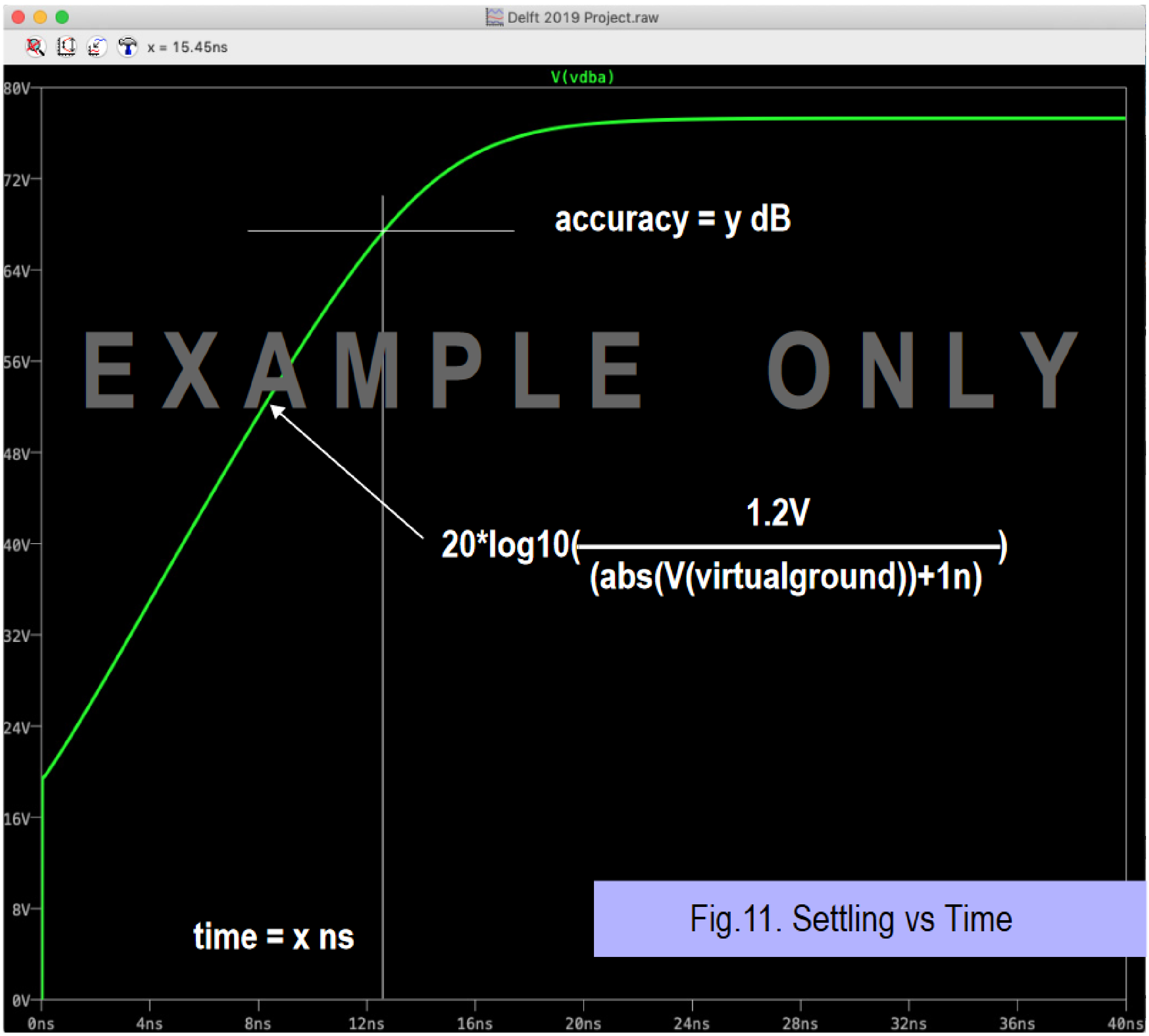

Fig.11. Settling vs Time

Example of deliverable 7:
Settling accuracy versus time with estimated time-constant τ_{cl}

In order to determine the closed-loop settling-time constant τ_{cl} from the settling curve (as in deliverable 6), you need to indicate the settling time T_{settle}, the time to reach 40 dB (T_{40dB}) and the time to reach 48.69 dB ($T_{48.69dB}$). The closed-loop settling time constant τ_{cl} can be calculated from this as the difference between the 2 values. From this result you can also calculate the closed-loop bandwidth BW_{cl}. Please compare this with earlier results.

As shown in the figure below, the settling part of the curve can be approximated by a straight line and the slope of this line can be used to determine the closed-loop bandwidth. There are, however, deviations from this straight line, both at the beginning (due to slewing) as well as when it starts reaching its maximum gain. To avoid those effects from influencing the estimation of the time-constant, you need to take the settling times at 40 dB and 48.69 dB accuracy. The 40 dB mark is safely past the slewing phase and the 48.69 dB mark is safely before the system reaches its ultimate gain. **Please compare the τ_{cl} you obtain here to the values found in the AC open-loop and AC closed-loop simulations.**

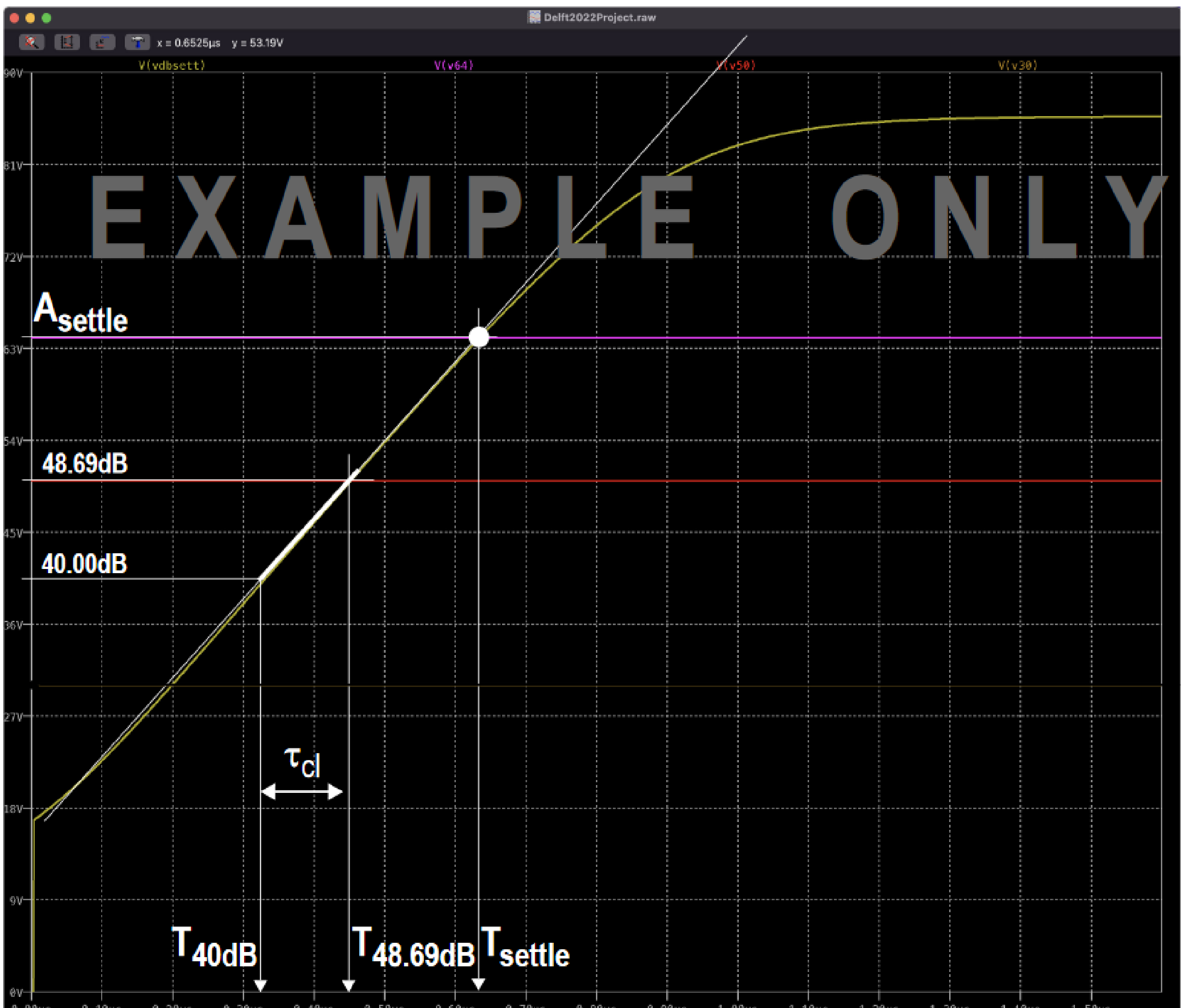

Fig.12. Settling curve with times indicated.

Example of deliverable 8:
Plot of output noise power density versus frequency

The plot of output noise density (Onoise) versus frequency allows you to integrate the total noise at the output. Please integrate the total noise from 10 kHz to 100 GHz. Be aware, this is not the same frequency range is in your AC-simulations. The **Total Integrated Noise** forms the lower end of your dynamic range or SNR. **The shown numbers are *examples* only and not a goal by itself (including the frequency range shown)!**

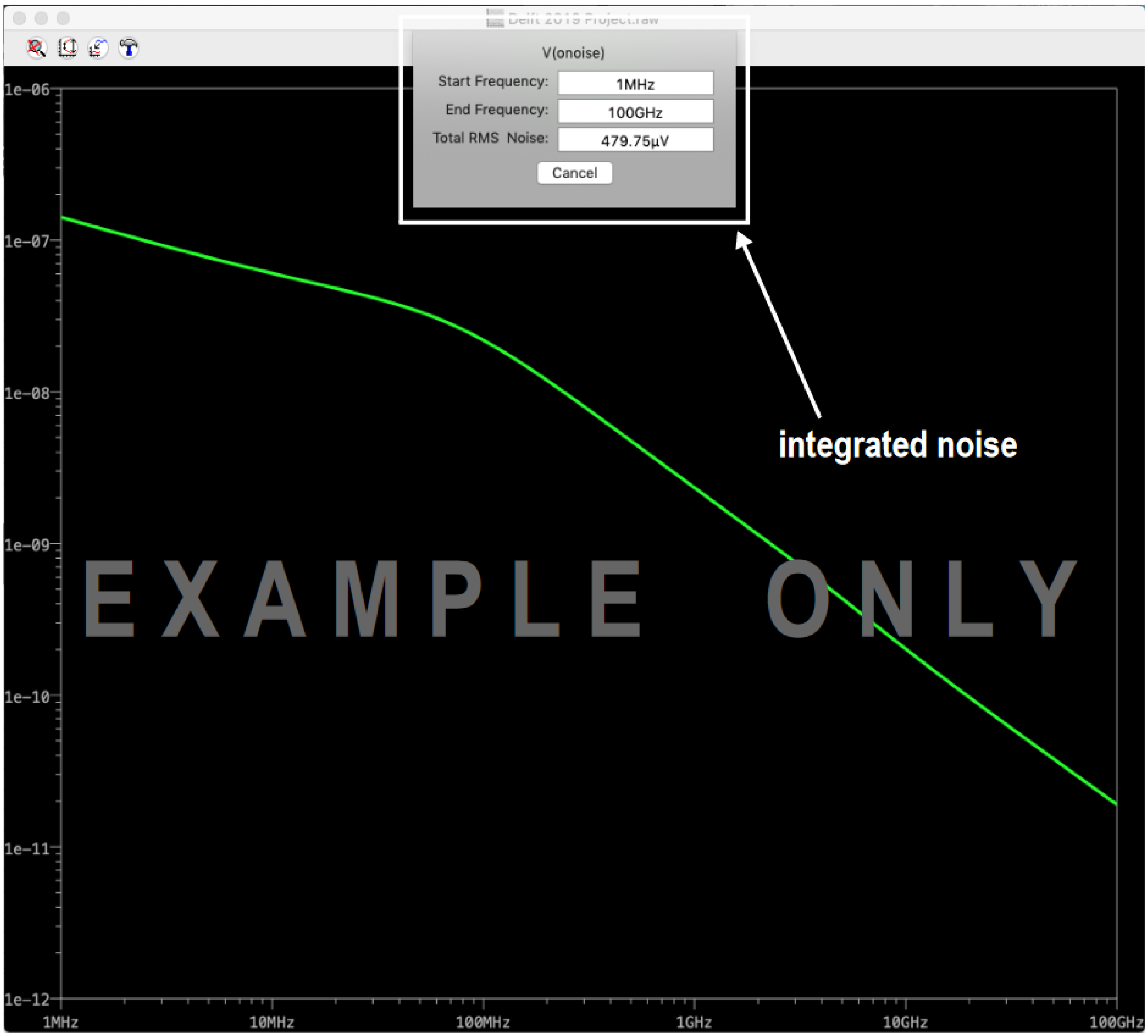

Fig.13. Onoise vs frequency and integrated noise shown.

COMPULSARY REPORT STRUCTURE

Page 1:	Title Page	Name + Student Number + EE4520 2024 / 2025
Page 2:	Deliverable 1	Table of Results
Page 3:	Deliverable 2	Schematic with node voltages
Page 4:	Deliverable 3	Schematic with branch currents
Page 5:	Deliverable 4	Bode-Diagram of A and Aß
Page 6:	Deliverable 5	Bode-Diagram of closed-loop gain
Page 7:	Deliverable 6	Plot of Settling behavior
Page 8:	Deliverable 7	Plot of Settling behavior with T_{settle}, T_{40dB} and $T_{48.69dB}$ indicated
Page 9:	Deliverable 8	Plot of Noise Density showing integrated noise
Page 10:	Deliverable 9	Short Explanation of the design process (max. 1 page)

Important note: a significant part of your HW2 grade will be based on the design choice explained in Deliverable 9. So, please make an effort to have a clear description.